

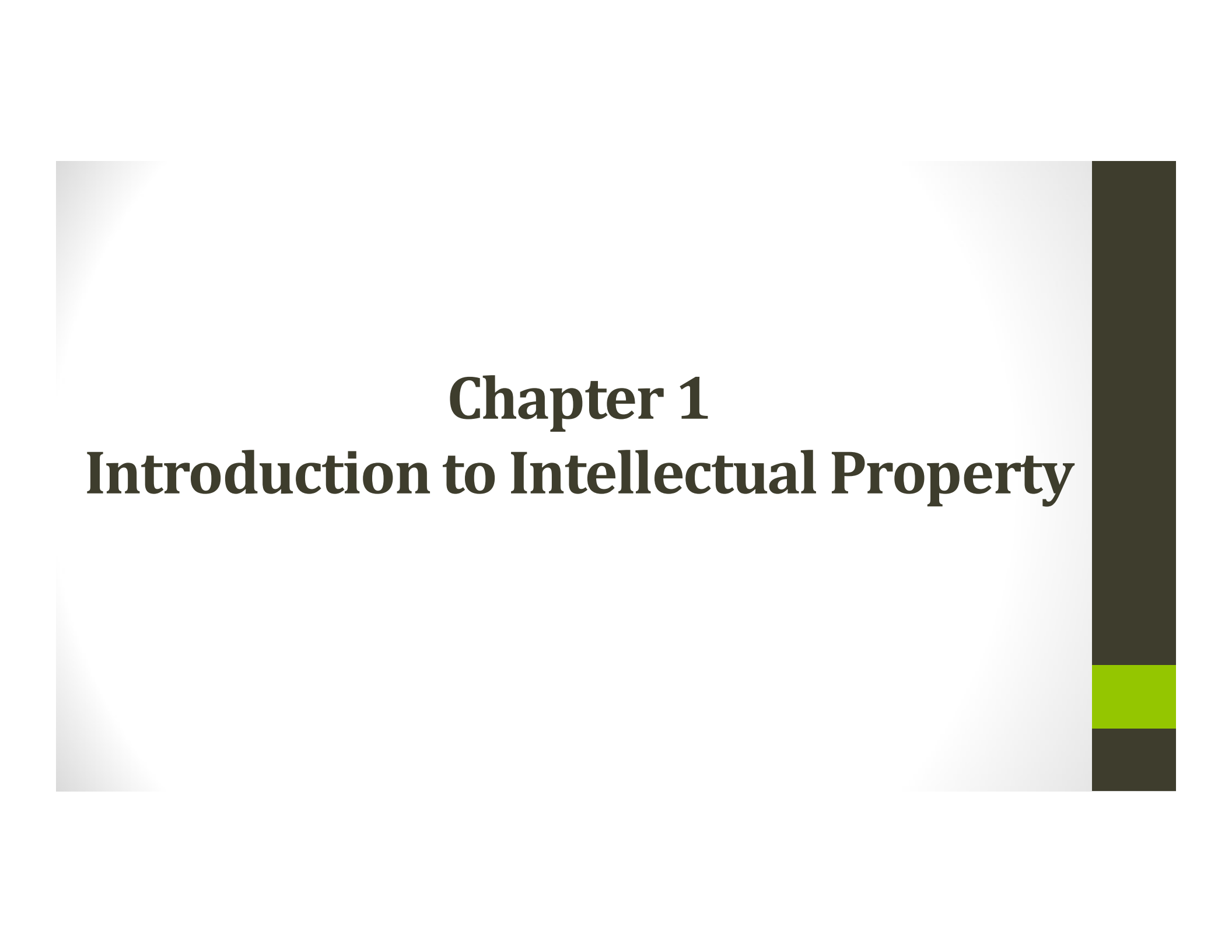
Module 3

Module-3(5 Hours)

Introduction To Intellectual Property: Role of IP in the Economic and Cultural Development of the Society, IP Governance, IP as a Global Indicator of Innovation, Origin of IP History of IP in India. Major Amendments in IP Laws and Acts in India.

Patents: Conditions for Obtaining a Patent Protection, To Patent or Not to Patent an Invention. Rights Associated with Patents. Enforcement of Patent Rights. Inventions Eligible for Patenting. Non-Patentable Matters. Patent Infringements. Avoid Public Disclosure of an Invention before Patenting. Process of Patenting.

Process of Patenting. Prior Art Search. Choice of Application to be Filed. Patent Application Forms. Jurisdiction of Filing Patent Application. Publication. Pre-grant Opposition. Examination. Grant of a Patent. Validity of Patent Protection. Post-grant Opposition. Commercialization of a Patent. Need for a Patent Attorney/Agent. Can a Worldwide Patent be Obtained? Do I Need First to File a Patent in India? Patent Related Forms. Fee Structure. Types of Patent Applications. Commonly Used Terms in Patenting. National Bodies Dealing with Patent Affairs. Utility Models.



Chapter 1

Introduction to Intellectual Property

INTRODUCTION TO INTELLECTUAL PROPERTY

- **Intellectual Property (IP)** is **any creation of the mind**, like inventions, designs, artwork, brand names, or symbols. It's similar to owning a physical object, but instead, **it's ownership of ideas and creative works**.
- **Intellectual Property Rights (IPR)** are the **legal rights that protect these creations**. **IPR gives** the creator or owner **control over how their IP is used, copied, or shared, often allowing them to earn money from it**. These rights help **creators feel secure** about **investing time and effort into their ideas**.

Example of IP: Imagine you invent a new smartphone app that helps people organize their daily tasks more efficiently. This app, as a creative product of your ideas and work, is considered your **Intellectual Property**.

Example of IPR: To protect your app, you can apply for a **patent** if the app includes a unique process or technology. Once granted, the **Intellectual Property Right (IPR)** gives you control over the app. This means no one else can legally use or sell your invention without your permission for a certain period.

Example of IP: Imagine you write an original song, including the lyrics and melody. This song is your **Intellectual Property** because it's a unique creation that came from your ideas and effort.

Example of IPR: To protect your song, you can apply for **copyright**. This **Intellectual Property Right (IPR)** gives you the exclusive right to decide who can copy, perform, or distribute your song. If someone wants to use it, they need your permission, and you can earn royalties each time it's played publicly or used commercially.

Example of IP:

The **iPhone** itself is a product that represents Apple's **Intellectual Property**. It combines innovative designs, unique software (iOS), hardware features, and technological processes that make it distinct from other smartphones.

Example of IPR:

- **Patents:** Apple holds **patents** on various aspects of the iPhone, such as its touchscreen technology, facial recognition features, and even the design of certain components. These patents give Apple the exclusive rights to use these technologies and prevent other companies from copying them.
- **Trademarks:** Apple also holds **trademarks** for the brand name "iPhone" and its logo (the apple with a bite taken out), which protects its brand identity. No other company can use the name "iPhone" or a similar logo for their own phones.

- **Two Main Branches of IP:**

- 1) **Copyrights and Related Rights:** refer to the **creative expressions** in the **fields of literature and art**, such as **books, publications, architecture, music, wood/stone carvings, pictures, portrays sculptures, films and computer-based software's/databases**.
- 2) **Industrial Property Rights:** These protect things like **inventions (patents), logos (trademarks), and designs** used in manufacturing or commerce.

- **Copyright:**

Copyright gives the creator the **right to control how their work (like a book, song, or software) is published, sold, or distributed**. It protects the creator's right to be recognized as the author of their work and stop others from using it without permission.

- **Patents:**

A **patent** is a legal right that gives the creator of an invention **exclusive control over the invention for a set period** (usually 20 years). **No one else can make, use, or sell the invention during that time** unless they get permission from the patent holder.

- **Trademarks:**

A **trademark** is a **symbol, word, or sign that distinguishes a company's products or services from others**. For example, the Nike "swoosh" logo is a trademark that makes it easy for people to identify Nike products.

- **Trade services**

It refers to **various financial and logistical services provided by banks or financial institutions** that **facilitate international trade and commerce**. These services **help businesses manage the complexities involved in buying and selling goods across borders**.

Trade services are **designed to streamline the international trade process, reduce risks, and ensure that transactions are conducted smoothly and efficiently**.



Example:

Let's say a business in India (the **seller**) is selling goods to a buyer in the United States (the **buyer**). To ensure the seller gets paid and the buyer receives the goods as agreed, the buyer's bank issues a **Letter of Credit** to the seller's bank.

Here's how the **Letter of Credit (LC)** works as a trade service:

1. The buyer requests their bank (the **issuing bank**) to open a letter of credit, guaranteeing payment to the seller once the seller meets the specified conditions (e.g., shipping the goods).
2. The issuing bank sends the letter of credit to the seller's bank (the **advising bank**), which informs the seller that payment is guaranteed, provided they fulfill the conditions.
3. The seller ships the goods and presents the required documents (like a bill of lading) to their bank.
4. The seller's bank checks the documents and ensures they match the terms in the Letter of Credit. If everything is in order, the bank transfers the payment from the buyer to the seller.
5. The goods are released to the buyer once the payment process is complete, and the transaction is secure for both parties.

- **Industrial Designs:**

Industrial designs **protect the unique appearance of a product**. For example, the shape or decoration of a product (like a chair or a bottle design) can be protected so that others can't copy it.

- **Geographical Indications:**

A geographical indication (GI) is a **name or sign used on products which corresponds to a specific geographical location or origin**. Items that meet geographical origin and quality standards may be endorsed with a government-issued stamp which acts as official certification of the origins and standards of the product.

Darjeeling Tea (India):

- Tea grown in the Darjeeling region of India is considered unique due to the region's specific climate and altitude. Darjeeling tea has a distinct flavor and aroma, making it highly prized worldwide.

Basmati Rice (India/Pakistan):

- Basmati rice grown in certain areas of India and Pakistan is protected under GI laws. The long grains and fragrant aroma of basmati rice are attributed to the soil and climate of the region.

Parmigiano Reggiano (Italy):

- This famous Italian cheese, also known as Parmesan, can only be produced in certain regions of Italy (Emilia-Romagna and parts of Lombardy) following traditional methods. The geographical region and the specific grass and feed available to cows contribute to its unique flavor.

ROLE OF IP IN THE ECONOMIC AND CULTURAL DEVELOPMENT OF THE SOCIETY

- **Creativity drives innovation, helping society to develop both socially and economically.** When **people create new ideas, technologies, or art**, it **pushes society forward**.
- **IPR gives creators legal rights to their work**, which **acts as a reward, motivating them to keep creating**. For instance, an author earns royalties from a book due to copyright, encouraging them to write more.
- **Strict IPR rules can sometimes harm people** by making essential products, like life-saving medicines, too expensive. When only one company has the right to make and sell a new product, they can set high prices, making it hard for many people to afford it. This can be especially harmful for essential items, as those who need them the most may not be able to buy them.
- **To balance creator rights with society's needs, exceptions to IPR exist.** For instance, the government allows certain uses of intellectual property to benefit the community.

One example of an IPR exception that benefits society is compulsory licensing for medicines. In certain situations, like a health emergency, the **government can allow other companies to produce a patented drug without the patent holder's permission.** This helps make the medicine more affordable and accessible to more people, especially in urgent cases. This exception ensures that while the creator's rights are respected, public health needs can be met quickly.

- An example of **protection from infringement**(infringement refers to using someone else's work (like a patented invention, copyrighted material, or trademark) without permission.) **claims** is when a teacher uses a copyrighted image in a classroom presentation without knowing it is protected by copyright. If the teacher uses it for educational purposes and is unaware of the copyright, the law might protect them from being sued(**Sued** means taking someone to court in a legal action.) for infringement, as long as it falls under fair use or similar exceptions. This **ensures that people can share knowledge and use creative works in good faith without fear of legal action.**
- The **use of copyrighted material for education and religious ceremonies** is **exempted** from the **operation of the rights granted in the Copyright Act.**

- **Compulsory Licensing in Emergencies**

During emergencies like natural disasters, **the government can revoke patents, allowing others to produce essential goods**. For instance, in a health crisis, **drug patents may be lifted to produce more medicines**.

- **No IP Rights for Harmful Creations**

If an **invention could harm society, the government won't grant its IP protection**. For example, cloning human embryos or creating dangerous viruses isn't allowed under IP laws because it could threaten public safety.

IP Governance

- Since IP is an integral component of human society, each and every nation has dedicated **agencies** for **laying out the guidelines, implementation and enforcement of IP related matters**.
- In India, many organizations/agencies deal with various aspects of IP. The **governance of all categories of IP is carried out by the Department for Promotion of Industry & Internal Trade (DPIIT) under the aegis of Ministry of Commerce and Industry, Govt. of India**.
- There are a few other dedicated organizations/departments **established by the government to promote patent-ecosystem (patent awareness, patent filing and patent commercialization) in India e.g. Technology Information Forecasting and Assessment Council (TIFAC), National Research Development Corporation (NRDC) and Cell for IPR Promotion and Management (CIPAM), etc.**
- For **smooth exchange of IP knowledge and products globally**, countries need to follow some common IP rules. This ensures fair treatment, like **similar patent protections, across countries**.

- The **UN** created **WIPO(World Intellectual Property Organization)** to make and share IP rules worldwide. WIPO helps countries follow standard rules for IP and helps people file IP internationally.
- WIPO oversees international treaties to protect IP globally. For example:
 - The **Paris Convention** protects patents and trademarks across member countries.
 - The **Patent Cooperation Treaty (PCT)** lets inventors apply for patents in multiple countries through a single application.
 - The **Berne Convention** protects authors' works worldwide, so a book copyrighted in one country is also protected in others.

IP AS A GLOBAL INDICATOR OF INNOVATION

- **IP**, especially **patents**, is considered as **one of the important parameter in assessing the innovation index of a nation**.
- The **global ranking organizations** always have **IP or a subset of IP** as **one of the parameters** for **understanding and grading the Science, Technology and Innovation (STI) ecosystem** of a nation.
- For example, the **Scimago** (publically available online portal which ranks journals and countries based on the data taken from Scopus) **2020 report** ranked **India at 4th position in the parameter of a number of Research Publications** and **50th position in the parameter of Intellectual Property Rights**.
- The **global ranking can be improved** by **sensitizing the teaching and scientific communities about the importance of IP** and **creating infrastructure for the same in the institutes of higher learning**.

ORIGIN OF IP

- **Early Origins of Intellectual Property (IP)**

The exact start of IP isn't known, but an early form of it **existed around 500 BCE in Sybaris, a Greek state**. People there were given **a year of protection if they invented something luxurious**. For example, if someone invented a **new type of comfortable furniture, others couldn't copy it for a year**.

- **IP in Medieval Europe**

A more organized way of managing IP started in medieval Europe.

In 1623, Britain introduced a law allowing guilds (groups of artisans or merchants) **to create new products and sell them**. For instance, **a guild of candle makers could make a unique candle and sell it without others copying it right away**.

- **The Statute of Monopolies**

This law replaced the earlier IP law due to public complaints. It gave the **rights to the original inventor for 14 years, meaning no one else could copy their invention during this period.** For example, **if someone invented a new tool, they would have 14 years to benefit from it before others could make the same tool.**

- **The Statute of Anne**

In 1710, Britain introduced the Statute of Anne, **giving authors rights to control how their books were copied and shared.** Authors could have **these rights for 14 years, and renew them for another 14 years.** For example, **a writer could control how their book was printed and sold.**

- **IP Laws Across Countries**

By the late 18th and early 19th centuries, many countries started making their own **IP laws to protect new inventions and creative works.** For instance, **inventors could secure patents for their creations in multiple countries, allowing them to benefit without others copying their work.**

HISTORY OF IP IN INDIA - patent

- The history of the **Indian patent system** dates back to the **pre independence era of British rule**.
- The **first patent related legislation in India** was **Act VI of 1856**, adapted from the **British Patent Law of 1852**. The objective of this legislation was **to encourage the inventions of new and useful manufactures**.
- At the **beginning of the 20th century**, introduced —**The Indian Patents and Designs Act 1911 (Act of 1911)**. As per this Act, **the governance of patents was placed under the management of the Controller of Patents**.
- Later changes added important features, like:
 1. **Government Use**: The **government could use certain inventions without permission**.
 2. **Patent of Addition**: **Allowed minor improvements to existing patents without a new application**.
 3. **Extended Patent Term**: **Increased from 14 to 16 years**.
 4. **Application Process**: **Allowed inventors to file a temporary (provisional) application, with nine months to submit a full application**

- **1949 Recommendations for National Interest**

In 1949, recommendations suggested changes to support public welfare:

1. **Prevent Misuse of Patents: Stopping patent owners from blocking essential uses.**

“imagine a patented **water purification technology** that could **help communities without clean water**. If the **patent owner refuses to allow others to produce or use this technology**, it could hurt public welfare. **Laws could step in to ensure that this technology is accessible, allowing people to benefit from it, while still respecting the patent owner's rights.**”

2. **Affordable Essentials: Ensuring affordable prices for food, medicine, and medical devices, while giving patent owners fair payment.**

“think of a **patented medicine needed for a widespread illness**. If the patent holder charges very high prices, many people might not be able to afford it. To ensure the public has access, the **government might set rules to keep prices reasonable, while making sure the patent owner is fairly compensated for their invention.**”

3. Law Updates: **Modifying parts of the Act to match the UK Patent Act.**

- **Act XXXII(32) of 1950: Implementing Changes**

These **1949 recommendations were put into law in 1950, adding protections for public access and fair prices.**

- **1952 Amendment for Compulsory Licensing (Act LXX of 1952)**
In 1952, an **amendment allowed compulsory licenses for essential products like food, drugs, and pesticides.** **This meant others could produce these items if patent owners weren't meeting public needs affordably.**

“A good example of compulsory licensing is when a life-saving drug is too expensive for most people to afford because only one company holds the patent. Under compulsory licensing, the government can allow other companies to produce cheaper versions of the drug, even without the patent holder's permission, if the original company is not meeting public needs. This makes the drug more accessible and affordable, ensuring that people who need it can get it.”

- **1957 Committee Report**

In 1957, a committee reviewed patent laws and suggested further updates. It addressed general issues with patent laws and discussed the rejected 1953 bill.

- **Patents Act, 1970**

In 1965, a revised patent law was submitted, which eventually became the Patents Act, 1970 after several revisions. This law is a foundation of India's patent system.

- **1999 Amendment: Product Patents for Pharmaceuticals and Agriculture**

In 1999, India allowed patents specifically for products like drugs and chemicals used in farming, known as "product patents." This law also created Exclusive Market Rights (EMRs), letting companies sell these products exclusively if certain conditions were met.

“For example, a pharmaceutical company developed a new, life-saving medication for heart disease. With the product patent law, the company could protect its formula, ensuring that no other company could make or sell the drug without permission. If the company met certain regulatory conditions, it could also secure Exclusive Market Rights (EMRs), giving it exclusive rights to sell the medication in India for a specified period, thus rewarding its innovation and covering research costs.”

- **Patents Act, 2002**

A new amendment in 2002 **updated the patent rules, creating a new version in 2003**. It **replaced the old rules from 1972 and modernized India's patent system**.

- **2005 Amendment Highlights**

With global changes in intellectual property, **another update came in 2005**. This amendment:

1. **Product Patents for All Technologies:** **Extended patent rights to inventions in all technology fields.**
2. **Preventing "Evergreening":** **Stopped companies from getting endless patents on minor changes to the same product** (like changing packaging for the same medicine).
3. **Rationalized Opposition Process:** **Improved how patents are challenged**, adding:
 - **Pre-Grant Opposition:** Others could challenge a patent application before it was granted.
 - **Post-Grant Opposition:** Challenges after a patent was granted.
4. **Compulsory License for Export:** **Allowed manufacturing of patented products** specifically for **export to other countries**.
5. **Compulsory License for Local Manufacture:** **Required patent owners to allow others to manufacture if the invention wasn't affordable or widely available.**
6. **Extended Grace Period for Exhibitions:** If a **product was shown in a government exhibition**, **inventors had 12 months (instead of 6) to file a patent.**

HISTORY OF IP IN INDIA - copyrights and related rights

- **Origins of Copyrights**

Copyrights go back to the **15th century but became important only after printing presses were invented**. Before printing, each copy of a book had to be handwritten, which was slow and prone to errors.

- **Evolution of Copyright Law in India**

In India, copyright laws evolved in three stages, mainly **during British rule and after independence**.

- **First Phase: The 1847 Copyright Law**

The **first copyright law in India** was introduced in **1847** by the **East India Company**. It **protected the author's work for their lifetime and for seven years after they passed away**. Copyright registration was required to claim these rights. **After an author's death, if the copyright holder refused to publish the work, the government could allow someone else to publish it.**

Example: If a book's author passed away and the family didn't want to publish it, the government could allow a publisher to print it to benefit the public.

- **Second Phase: The Copyright Act of 1914**

In 1914, the **British updated India's copyright laws to align with the UK's law from 1911**. This law introduced **penalties for those who copied work illegally**.

Example: If someone printed copies of a book without permission, they could be fined or face legal action.

- **Third Phase: The Copyright Act of 1957**

After independence, India created the Copyright Act of 1957, **replacing the older laws to meet international standards**, specifically the **Berne Convention**, which **protects authors' rights worldwide**.

Example: An Indian author's work is protected in other countries that follow the Berne Convention, ensuring their rights are respected internationally.

- **Amendments to the 1957 Act**

India's copyright law has been **updated six times** (in 1983, 1984, 1992, 1994, 1999, and 2012) to **meet new global copyright agreements**, such as the **WIPO Copyright Treaty and the WIPO Performances and Phonograms Treaty, which focus on digital rights and performances**.

Example: These updates help protect Indian movies and music globally, especially in digital formats like streaming.

- **India's Participation in International Copyright Treaties**

India is part of many global copyright treaties, like the **Berne Convention (which protects works internationally)** and the **TRIPS Agreement (which protects IP across trade agreements)**.

Example: A book published in India is also protected in other countries that are part of the Berne Convention, making it harder for unauthorized copies to be sold abroad.

HISTORY OF IP IN INDIA - trademarks

- **First Trademark Law in India (1940)**

India's first law for trademarks was the Trade Marks Act of 1940. It was adapted from the UK's Trade Marks Act of 1938. This law helped protect brand names, logos, and other marks that identify a business or product.

- **Trademark Rules Added to Other Laws**

Rules for trademarks were later included in other Indian laws, like the Indian Penal Code and the Sea Customs Act, to make it easier to handle trademark issues in various situations.

“when someone tries to import counterfeit products with fake brand logos, like fake "Nike" shoes. Under the Sea Customs Act, customs officers can now stop these counterfeit items from entering the country, as the trademark rules are included in this law. This prevents fake goods from flooding the market, protecting both consumers from poor-quality products and companies from brand misuse. Additionally, under the Indian Penal Code, selling products with fake trademarks is a punishable offense, helping protect genuine brands and their reputations.”

- **Trade Marks Act of 1958**

The Trade Marks Act of 1940 was renamed the Trade and Merchandise Marks Act in 1958. This updated law **aimed to protect both brand names and other kinds of marks used in business.**

“An example of the Trade and Merchandise Marks Act protecting brand names is when a **popular soft drink company registers its logo and name as trademarks.** This **prevents other companies from using similar names, logos, or packaging** that might confuse customers into thinking they’re buying the well-known brand. If a new company tries to use a similar logo or name to sell its drinks, the original brand can take legal action to protect its identity and prevent customers from being misled. This **Act helps businesses maintain a unique brand identity, protecting them from imitators and ensuring consumers know they’re getting the original product.”**

- **Trade Marks Act of 1999**

In 1999, the old Act was replaced by a new Trade Marks Act **to follow the international TRIPS agreement, which sets global standards for trademark laws**. This 1999 Act is the law that **India still uses today to register and protect trademarks**.

“An example of the Trade Marks Act of 1999 in action is how a **company like Tata registers its logo as a trademark in India**. This trademark **protects Tata’s brand identity**, meaning other companies cannot legally use a similar logo or name to confuse customers. Under this Act, if someone tries to use the Tata logo or a name that could mislead customers into thinking it’s related to Tata, the company can take legal action to stop them. **This law aligns with international standards, helping Indian businesses protect their brands both in India and globally.**”

HISTORY OF IP IN INDIA - geographical indications

- India, as a **member of WTO**, enacted the **Geographical Indications of Goods (Registration and Protection) Act, 1999**.
- It came into force with effect from **15th September 2003**. **Geographical Indicators** have been **defined under Article 22 (1) of the WTO Agreement on TRIPS**. [Trade-Related Aspects of Intellectual Property Rights.]

“What is a Geographical Indication (GI)?

A GI is a sign that shows **a product comes from a specific location and has qualities or reputation linked to that place**. For example, Darjeeling tea is known for its unique flavor and can only be called "Darjeeling tea" if it is grown in that region of India.”

HISTORY OF IP IN INDIA - Industrial Design

- **Need for Protecting Industrial Designs**

In the **18th century**, people saw the **need to protect the unique designs used in products to recognize the work of designers and inventors.** In India, the first law for this was the **Patterns and Designs Act of 1872**, which **aimed to protect creators' rights over their new patterns and designs.**

Example: If someone created a new pattern for fabric, this law helped protect it so others couldn't copy it without permission.

- **British Patents and Designs Act of 1907**

The 1872 Act was later **replaced by the British Patents and Designs Act in 1907.** This Act formed the **foundation for future Indian design laws, eventually leading to the Indian Patents and Designs Act of 1911.**

Example: A unique furniture design could be protected under this law, allowing only the designer to produce or sell it.

- **Patent Act in 1970**

In 1970, India created a new, separate law for patents called the *Patent Act, 1970*, which focused only on inventions (like new machines or processes). The *Indian Patents and Designs Act of 1911* was then kept only for protecting designs.

Example: While patents protected inventions like a new type of engine, designs still needed protection for things like the shape of a product.

- **Industrial Designs Act of 2000**

Finally, in 2000, India passed a specific law just for industrial designs called the *Industrial Designs Act*, which officially took effect in 2001. This law focuses only on protecting the visual appearance of products.

Example: This Act protects the design of items like a smartphone's shape or a unique bottle design, so companies can't copy each other's look and feel.

Example:-

A practical example of the **Industrial Designs Act of 2000** is the protection of Coca-Cola's uniquely shaped glass bottle. This distinctive bottle design is protected under industrial design law, meaning that other beverage companies cannot legally produce bottles with a similar shape that could confuse consumers. The Act ensures that Coca-Cola retains exclusive rights to this specific bottle appearance, protecting its brand identity and allowing it to stand out in the market. This kind of design protection helps companies maintain a unique look for their products, giving them a competitive edge.



History of IP in India - Semiconductor Integrated Circuits and Layout designs

- In the **21st century**, **Information Technology (IT)** has revolutionized the **economic and societal growth of the world economy**.
- The rapid and tremendous scientific advancements in the field of IT resulted in the creation of a new class of IP called the **Layout-Design of the Semiconductor Integrated Circuits**.
- Various organizations, including **WTO and TRIPS Agreement laid down rules and regulations regarding the protection** of Semiconductor Integrated Circuits and Layout Designs (**SICLD**)
- As a WTO member, India passed the **Semiconductor Integrated Circuits Layout-Design (SICLD) Act in 2000**. This law **aligns with TRIPS rules (Articles 35–38)** and protects semiconductor layout designs in India, just like in other countries.
- <https://sicldr.gov.in/sicw/index>

History of IP in India - Plant varieties

- **Early Patent Focus on Microbes, Not Plants or Animals**

Before the 1970s, patents mostly covered inventions involving microbes (like bacteria) but didn't focus much on plants or animals. For example, creating a new type of bacteria for medicine could be patented, but a new plant variety could not.

Example:- According to India's Patents Act, a scientist who creates a new bacteria that cleans oil spills can patent it. But a breeder who creates a new plant variety cannot patent it under this Act, ensuring plants and animals remain common resources.

- **New Law to Cover All Biological Materials**

To make sure **plants and animals were also protected under patent laws**, a special type of law was created, called a **"sui generis" law**. This was set up under the **International Convention for the Protection of New Varieties of Plants (UPOV) in 1978 and updated in 1991**. These laws help **protect unique plant breeds that breeders develop**.

Example:- For example, under UPOV, a company that developed a drought-resistant wheat variety could apply for legal protection, giving them exclusive rights to sell that wheat variety.

- **Concern for the Environment and Public Interest**

These changes in law were made **to protect the environment and public interests**. For instance, **protecting new plant varieties can help preserve biodiversity, which benefits everyone**.

- **India's PPV&FR Act of 2001**

To follow international rules (**specifically Article 27.3(b) of the TRIPS Agreement**), India introduced the **Protection of Plant Varieties and Farmers' Rights (PPV&FR) Act in 2001**. This **law not only protects new plant varieties developed by breeders but also secures farmers' rights to save, share, and reuse seeds**.

Example:- if a company develops a high-yield rice variety, they can protect it under this law. However, farmers are also given rights to save seeds from this rice and share them with neighbors, ensuring that they are not forced to buy seeds every season

What is Biodiversity?



Biodiversity is the variety of all living things on Earth, like plants, animals, and even tiny insects. It helps keep nature balanced and makes our planet a healthy place to live.

Biodiversity conservation

- In 1927 the —Indian Forest Act and later on the —Wildlife Protection Act 1972 was enacted to provide legal protection to biodiversity.
- In 1988, the —National Forest Policy was passed, which brought revolutionary changes in the conservation and management of biodiversity.
- The Acts and policies in force to protect the environment and biodiversity in India include Mining and Mineral Development Regulation Act, 1957; Water (prevention and control of pollution) Act, 1974; Forest Conservation Act, 1980; Biological Diversity Act, 2002; Scheduled Tribes and other Traditional Forest Dwellers (recognition of rights) Act, 2006; National Biodiversity Action Plan, 2009; National Environment Policy, 2006 and a few more.

Major amendments in IP laws and acts in India

.No.	Year	Historical Proceedings
Patents		
1.	1856	The Act VI of 1856 on the protection of inventions based on the British Patent Law of 1852.
2.	1859	➤ Rights renamed as 'Exclusive Privileges'. ➤ Time for the priority increased from 6 months to 12 months.
3.	1883	➤ The Patterns and Designs Protection Act ➤ Introduction of novelty in the invention. ➤ A grace period of 6 months for the disclosure of the invention.
4.	1911	➤ Renamed as 'The Indian Patent and Design Act' and brought under the management of Controller of Patents'.
5.	1930	➤ Introduction of Patent of Addition. ➤ Government can use the invention if required. ➤ The term of patent protection increased from 14 to 16 years.
6.	1945	➤ Filing of the provisional specification to secure the priority date. ➤ Provision of submitting complete specifications within 9 months.
7.	1949	➤ Dedicated Committee formed under the leadership of Justice Bakshi Tek Chand for reviewing patent system as per the national environment.

8.	1950	➤ A working statement needs to be submitted at the Patent Office. ➤ Endorsement of the Patents with the words 'License of Right' on the application made by the government so that the Controller could grant the license.
9.	1952	➤ Provision of 'Compulsory License' in the areas of food, medicine and insecticide germicide. ➤ Process for producing substance or any invention relating to surgical or curative devices.
10.	1965	➤ After incorporation of the recommendation submitted by the committee formed in 1949, a new bill was introduced in Lok Sabha but was not cleared.
11.	1967	➤ Again submitted to Parliamentary Committee. ➤ 1911 Act remained applicable for Designs.
12.	1970	➤ The Patent Act, 1970 passed by the Parliament Committee.
13.	1972	➤ The Patent Act, 1970 came into force with the introduction of patent rules.
14.	1995	➤ TRIPS Agreement was signed by India and got transition period 1995-2005 to make domestic laws compatible with TRIPS.
15.	1999	➤ Introducing the provisions for receiving the applications for the product patent in the field of pharmaceuticals and agro-chemicals (mail box)*. ➤ Provisions for the grant of EMRs for distribution and sale of pharma products on fulfilment of certain conditions. ➤ Grant of EMR subject to certain conditions. * after the amendments (1999) the product
		➤ patents related to the pharmaceuticals and agro- chemicals were kept on hold for examination till 2005. It is called a mailbox or black box.
16.	2002	➤ The uniform 20-year term of the patent for all inventions. ➤ Disclosure of source and geographical origin of biological material made compulsory. ➤ Establishment of Appellate Board. ➤ Compulsory License provisions strengthened.
17.	2003	➤ The Patents Rules, 2003 were introduced.

18.	2005	➤ Product patent for inventions in all fields of technology including food, drug, chemicals and microorganisms. ➤ New forms of known substances excluded in order to prevent the ever-greening of the patent. ➤ Introduction of the pre-grant opposition. ➤ Introduction of post-grant opposition. ➤ Extension of grace period to 12 months.
Copyrights and Related Rights		
19.	1847	➤ The concept of Copyrights in India was introduced. ➤ Validity - Lifetime+7 years but not more than 42 years in total.
20.	1914	➤ Copyright Act, 1914 was introduced based on the Imperial Copyright Act, 1911 of UK.
21.	1957	➤ Copyright Act, 1914 was replaced with Copyright Act, 1957 with minor modifications.
22.	1984	➤ Penalty on second and subsequent conviction.
23.	1994	➤ Registration of Copyright Society made mandatory.
24.	2012	➤ To comply with international Treaties for copyrights protection in the digital environment. ➤ Right to receive royalties for authors and music composers. ➤ Exception of copyrights for physically
		➤ To comply with international Treaties for copyrights protection in the digital environment. ➤ Right to receive royalties for authors and music composers. - Exception of copyrights for physically disabled persons to access any work.
25.	2013	➤ Copyrights Rules, 2013 introduced.
Trademarks		
26.	1940	➤ Trademarks Registry established in India.
27.	1958	➤ The Trade and Merchandise Marks Act, 1958 enacted as per TRIPS Agreement.
28.	1999	➤ Amended to avoid duplicity and ensure securing proprietors' trade and

		goodwill.
29.	2002	➤ Trademarks Rules introduced.
30.	2010	➤ Amended to comply with Madrid Protocol for international filing. ➤ Provision for filing opposition of theregistration within 4 months.
31.	2013	➤ Trademarks Rules introduced.
Geographical Indications		
32.	1999	➤ Being a member of the World Trade Organization (TRIPS), GI of goods (Registration and Protection) Act was introduced.
33.	2002	➤ The Geographical Indications of Goods (Registration and Protection) Rules, 2002 was introduced.
34.	2003	➤ The Geographical Indications of Goods (Registration & Protection) Act came into force.
Designs		
35.	1872	➤ Patterns and Designs Protection Act introduced for the protection of new patterns and designs.
36.	1888	➤ Amended as Invention and Design Act, 1888 for the protection of new inventions and designs.
37.	1911	➤ Renamed as The Indian Patent and Design Act.
38.	2000	➤ Design Act, 2000 was introduced; separated from the Indian Patent and Design Act.
39.	2001	➤ Design Rules, 2001 introduced.
Semiconductor Integrated Circuits Layout Design (SICLD)		
40.	2000	➤ Semiconductor Integrated Circuits Layout Design ➤ (SICLD) Act 2000 introduced as a signatory of WTO.
41.	2001	➤ SICLD Rules introduced.
Protection of Plant Varieties and Farmers' Rights (PPV&FR)		
42.	1970	➤ The Patent Act, 1970 excluded plants and animals in whole or in any part from patentability (in 1999 amendments).

43.	1991	➤ Enactment of protection of new varieties of plants on <i>sui generis</i> basis on the lines of UPOV.
44.	2001	➤ In line with TRIPS Agreement enactment of PPV&FR Act was introduced.
Biological Diversity		
45.	2002	➤ The Biological Diversity Act, 2002 introduced on the lines of the Convention on Biological Diversity (CBD, 1992).
46.	2003	➤ Establishment of National Biodiversity Authority. ➤ Designation of repositories under the Biological Diversity Act.
47.	2004	➤ Biological Diversity Rules introduced.

Chapter – 2

Patents

- A patent is a legal protection given to an inventor for a new idea or product.
- Think of it as a special license that allows only the inventor to make, use, or sell their invention.
- This protection is granted because the inventor has shared enough details so that others could, in theory, replicate the invention if the patent did not exist.
- In return, the inventor has the exclusive right to use or license the invention for a limited time.

Example: Imagine someone invents a new type of phone battery that charges in just 5 minutes. They apply for a patent to prevent other companies from copying or selling this battery without their permission. If they get the patent, they can control who makes or uses this battery technology for a certain period—usually 20 years. In their patent application, they must explain how this battery works in a way that a skilled engineer could understand. Once the patent expires, others can use the technology, but until then, the inventor has the exclusive right.

CONDITIONS FOR OBTAINING A PATENT PROTECTION

- There is a set criterion, as provided in Section 2(1)(j) of the Patents Act, 1970, which must be fulfilled for a product or a process to qualify for the grant of a patent.

1) **Novelty** - Not part of 'State of the Art'. The innovation claimed in the patent application is new and not known to anybody in the world.

In other words, the innovation is,

- not in the knowledge of the public,
- not published anywhere through any means of publication and
- not be claimed in any other specification by any other applicant.

2) **Inventive step** - Not obvious to the person (s) skilled in the art.

The innovation is

- a technical advancement over the existing knowledge,
- possesses economic significance and,
- not obvious to a person skilled in the concerned subject.

- **Example:** Imagine an engineer develops a mobile phone that can automatically adjust its volume based on the background noise level. If this is a new, useful idea that experts wouldn't naturally think of, it may meet the inventive step requirement.

3) **Capable of industrial application** - For the benefit of society. The invention is capable of being made or used in any industry.

Example: Suppose an inventor creates a machine that can quickly and efficiently convert plastic waste into a reusable building material. This machine can be manufactured and used in industries dealing with construction materials, making it useful on a large scale. Because the machine offers a practical, industrial solution to the problem of plastic waste, it meets the requirement for industrial application.

TO PATENT OR NOT TO PATENT AN INVENTION

- When an inventor creates something new, they have a choice: they can either use this invention for personal gain, following the legal protections in their country, or they can share it with the public so anyone can use it without needing to pay them.
- Typically, inventors choose to benefit personally from their inventions, and very few inventors give up their rights to allow free use by everyone.
- If an inventor decides to earn money from their invention, they have two main choices: they can either **patent it** or keep it as a **trade secret**.
- **Trade Secrete**:- This is like a hidden recipe that the inventor doesn't want anyone else to know. If the inventor believes they can keep it secret for a long time—like the formula for Coca-Cola, which has been kept secret for over a hundred years—then they might choose this option. Trade secrets are effective if the invention can stay private for many years, and there's a very low risk that someone could figure it out by examining the final product.
- **Trade Secret** = Keep it hidden for as long as possible (like a secret recipe).

- **Patent:** A patent is like a public agreement. You tell the government exactly how your invention works, and they give you legal protection. This means you're the only one who can use or sell that invention for a set time, usually 20 years. This is a good option if your invention could be easily copied or if it might only be useful for a short time (like a new technology that will be outdated in a few years). Once you have a patent, no one else can use your invention without your permission, allowing you to make money from it during that time.
- **Patent** = Share how it works publicly, but get legal protection so only you can use it for a limited time.

RIGHTS ASSOCIATED WITH PATENTS

- Patents give inventors exclusive control over their inventions, allowing them to decide who can use, make, or sell their creation. Imagine you invent a new type of eco-friendly battery. By getting a patent, you have legal protection, meaning:
 1. **Only you can decide who uses your battery idea:** Others can't make, sell, or import it without your permission.
 2. **You can allow others to use it on your terms:** You might allow a company to use your battery design if they pay you a fee. This agreement is called licensing.
 3. **Patent rights prevent others from using your idea without permission:** These rights are often called “negative rights” because they restrict others from using the invention without consent. So, without your approval, no one else can legally work with your design.
 4. **You can take legal action if someone uses it without permission:** If a company starts selling a similar eco-friendly battery without asking you, you can sue them. You can ask the court to stop their actions and even seek compensation.

ENFORCEMENT OF PATENT RIGHTS

- Enforcement is the process of ensuring compliance with laws, regulations, rules, standards and social norms.
- Patent rights are usually enforced by the judicial courts.
- The Court of Law has the authority to stop patent infringement.
- However, the main responsibility for monitoring, identifying and taking action against infringers of a patent lies with the patent owner.
- Enforcement is about making sure that people or organizations follow rules, laws, or standards. In the case of patents, enforcement means ensuring that no one uses an invention without permission from the patent holder (the person or organization that owns the patent).

Here's a simple example:

1. **Patent Rights Enforcement:** Suppose someone invents a new type of phone that can charge itself with sunlight. They file for a patent to protect their invention, so only they can produce or sell this phone for a certain period.
2. **Role of the Patent Owner:** If someone else starts making or selling similar phones without permission, it's up to the patent owner to notice this. The patent owner can then take legal action to stop the infringer.
3. **Court's Role in Enforcement:** If the patent owner decides to act, they can go to court. The court has the authority to investigate and, if the patent owner's rights are being violated, it can order the infringing company to stop making or selling the phones.

INVENTIONS ELIGIBLE FOR PATENTING

- A patent is a legal right granted to an inventor that protects their unique invention or improvement, preventing others from making, using, or selling it without permission. This applies to a wide range of things, from everyday objects to complex technologies.
1. **Inventions in Any Field:** Patents can be granted for inventions in any field, whether simple like a paper clip or something advanced like a nanotechnology chip. For example, a **Harvard mouse**, which is a mouse with cancer genes, can be patented.
 2. **Patents Aren't Just for Big Breakthroughs:** Many people think patents are only given for groundbreaking discoveries, but that's not true. In fact, most patents are for **improvements** on existing inventions. For example, improvements in the design of a pen or a better version of a medicine.
 3. **Example of Penicillin:** Penicillin, a well-known antibiotic, can have various **derivatives** (modified versions). For instance, by changing the chemical structure of penicillin slightly, new versions are created that may be stronger, work in different conditions, or target a wider range of bacteria. Each new version can get a separate patent. These are called **second-generation** or **third-generation** penicillins.

4. **Everyday Patented Items:** Many common products around us are protected by patents. Things like **toothbrushes**, **shoes**, **mobile phones**, **wristwatches**, and even **bicycles** involve patented technologies or designs.
5. **Products with Multiple Patents:** Some products, like a **laptop** or a **car**, might include several patents for different parts. For example, a laptop may have patents for its screen, keyboard, battery, processor, and other features. Similarly, mobile phones and cars have multiple patented components working together.

NON-PATENTABLE MATTERS

- The **Patent Act, 1970** defines certain inventions that cannot be patented. These are called **non-patentable matters**, meaning these inventions or ideas are excluded from getting patent protection.

1) Invention contrary to public morality:

These are inventions that go against societal values or morals.

Example: A method for **human cloning** or a new way to run **gambling operations** might not be allowed because they can raise ethical concerns.

2) Mere discovery:

If you just **find** something that already exists in nature or that is already known, you can't patent it.

Example: Finding a **new microorganism** in nature or discovering the **laws of gravity** can't be patented because they are natural phenomena, not inventions.

3) Mere discovery of a new form of a known substance:

If you find a new form of an existing substance but don't add anything new, it can't be patented.

Example: Suppose someone discovers that **Vitamin C** (ascorbic acid), which is well-known for boosting immunity, also has benefits for improving skin health. Even though this new use is valuable, it doesn't make Vitamin C itself a new invention. The substance is already known, and simply finding another benefit or form of its use doesn't qualify for a patent.

4) Frivolous invention:

- A **frivolous invention** is something that is not really useful or doesn't bring anything new to the table. It may seem like an invention at first, but it doesn't solve a real problem or offer something valuable.
- frivolous inventions are those that don't have practical use or don't offer something important or innovative.

Example: Imagine someone invents a **spoon with a built-in napkin holder**. While it might sound clever at first, it's really just a spoon with a napkin holder attached. It doesn't solve any big problem or add real value to the idea of a spoon. It's just a silly or unnecessary twist on something that already exists. Since it doesn't provide anything useful or new, it wouldn't be able to get a patent.

5) Arrangement or rearrangement:

If you just attach two things together without creating something truly new, you can't patent it.

Example: An **umbrella with a fan** or a **torch on a bucket** are just combinations of already existing items, so they don't qualify for a patent.

6) Inventions Related to Atomic Energy are excluded from patent protection because they involve materials or processes that are sensitive to national security, safety, and environmental concerns. This exclusion is primarily to prevent the misuse of inventions involving nuclear or radioactive substances and to ensure these powerful resources are managed and regulated by the government.

7) Literary, dramatic, musical, or artistic work:

These kinds of works are protected by **copyright law**, not patent law.

Example: Books, paintings, or **computer programs** can't be patented. They are copyrighted instead because they are creations of art or literature.

8) Topography of integrated circuits:

The layout or design of electronic circuits, like those in chips, is not patented under the Patent Act but is protected by a separate law.

Example: Circuit board designs for electronics are protected under the **Semiconductor Act**, not the Patent Act.

9) Traditional knowledge:

Inventions based on traditional knowledge, or those that simply repeat old methods, cannot be patented.

Example: If you create a product based on **ancient herbal remedies** without adding something new, it wouldn't be patentable because it is based on traditional knowledge that has already been passed down.

PATENT INFRINGEMENTS

- Patent infringement occurs when someone uses, makes, or sells an invention that is protected by a patent without getting permission from the patent owner. A patent is like a legal protection that gives the inventor exclusive rights to use or sell their creation. If others use the invention without permission, it is considered an infringement.
- There are two main types of patent infringements: direct and indirect.

1) Direct Infringement: This happens when someone knowingly uses, makes, or sells a product that is very similar to a patented product without the owner's permission.

- **Example:** If someone invents a new type of smartphone camera technology and receives a patent for it, no one else can make or sell phones with the same camera technology without permission. If a company copies this technology without authorization, it's a direct infringement.

2) Indirect Infringement: This occurs when someone unintentionally or accidentally infringes on a patent without the intention to copy or harm. It can also involve someone helping others to infringe, even if they didn't directly use the invention.

- **Example:** Suppose a company unknowingly sells a product that uses a patented component without realizing it's patented. Even though it was unintentional, it can still be considered an indirect infringement.

Legal Actions and Remedies

- When an infringement occurs, the patent owner has the right to take legal action against the infringer. The owner may seek:
- **Interim injunction:** A temporary order to stop the infringer from using or selling the patented invention until the court makes a final decision.
- **Damages or profits:** The patent owner can seek financial compensation for any losses or profits the infringer gained from the unauthorized use.
- **Permanent injunction:** A long-term or permanent order stopping the infringer from using the patented invention in the future.

AVOID PUBLIC DISCLOSURE OF AN INVENTION BEFORE PATENTING

- If you have an invention, it's best to keep it private until you apply for a patent. Once something is shared publicly—through a display, publication, or online—it usually can't be patented, because it loses its "novelty," or newness, which is a key requirement for patents.
- However, there are some exceptions. The Patents Act allows a 12-month "grace period." This means that if you publish your invention in a journal, present it at a scientific event, or display it at an exhibition, you still have 12 months to file a patent application without losing the chance to protect your invention.
- Sometimes, though, you might need to share your invention with someone before filing the patent, like with a potential investor or business partner. For example, if you have invented a new type of solar panel, a company might want to know exactly how it works to decide if they want to invest. In this case, to protect yourself, you should ask them to sign a Non-Disclosure Agreement (NDA), which legally prevents them from sharing your invention with others or using it without your permission.

Chapter – 3

Process of Patenting

PROCESS OF PATENTING

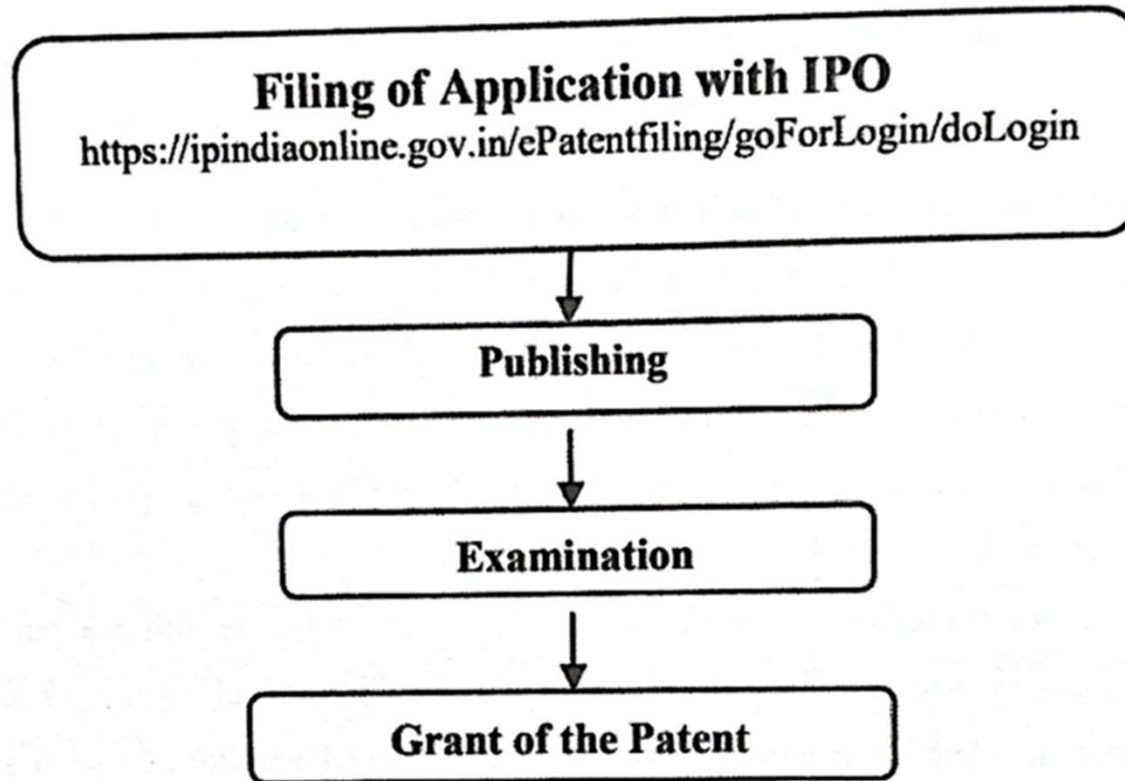


Figure 6.1: Flow chart of major steps involved in the grant of a patent

- In India, the process of grant of a patent is a lengthy procedure that may take anywhere 3- 4 years or more.

PRIOR ART SEARCH

- **Novelty Check for Patents:** Before filing a patent, inventors need to confirm that their invention is new and not already publicly known. They do this by checking if the invention exists in the public domain.
- For this, he/she needs to read patent documents and Non-Patent Literature (NPL), scientific journals/reports/magazines, etc.
- **Prior Art:** This term refers to any existing public information or data about an invention, regardless of format, that could affect its patent eligibility. This information must be known before the patent's filing or priority date.

- **Benefits of a Prior Art Search:** Checking for prior art before patent filing has key benefits:

1. **Prevents Infringement:** Avoids accidentally copying or infringing on existing patents.
2. **Supports Research and Development:** Helps understand current trends and research directions.
3. **Offers Detailed Knowledge:** Provides comprehensive information on similar inventions.

- **Search Criteria:** A prior art search typically assesses:

1. **Novelty** (Is the invention new?),
2. **Patentability** (Can it be patented?),
3. **State of the Art** (Current advancements in the field),
4. **Infringement Risk** (Does it conflict with existing patents?),
5. **Validity** (Is it likely to withstand legal challenges?),
6. **Freedom to Operate** (Can it be used without legal restrictions?).

- The patent databases that are available are:

Patent Databases

UNPAID

- Indian Patent Advanced Search System (InPASS) available at <http://ipindiaservices.gov.in/publicsearch/>
- Patentscope by WIPO accessible at <https://www.wipo.int/patentscope/en/>
- Espacenet provided by the EU, found at <https://worldwide.espacenet.com/patent/>
- USPTO of the USA, accessible via <https://www.uspto.gov/>
- Google Patents Advanced Search accessible at <https://patents.google.com/advanced>

PAID

- Orbit Intelligence available at <https://www.questel.com/business-intelligence-software/orbit-intelligence/>
- Derwent Innovation provided by Clarivate, available at <https://clarivate.com/derwent/solutions/derwent-innovation/>
- PROQUEST search at <https://about.proquest.com/search/?searchKeyword=patent+>

Non-Patent Literature (NPL)

- Scholarly publications including handbooks, textbooks, withdrawn patents, encyclopedias, journals (such as IEEE, Research Gate, Springer, Wiley Online Library, etc.), dissertations, NCBI's PubMed, conference proceedings, technical reports, and public conferences.
 - Industry/trade publications encompassing industry reviews, public disclosures (found on social media, YouTube, books, magazines, datasheets, blueprints, etc.).
 - Other sources like newspapers, websites, technology blogs, researchers' websites, etc.
- While much of the NPL data is accessible freely on public platforms, certain journals may require payment for access due to subscription models. Notably, major patent offices like the United States Patent and Trademark Office (USPTO), European Patent Office (EPO), Japan Patent Office (JPO), etc., maintain their own in-house NPL databases to enhance the efficiency of patent examination.

CHOICE OF APPLICATION TO BE FILED

- Once an inventor decides to patent an invention, they must choose between filing a provisional or a complete **(final) patent application**. A **provisional patent application** is often chosen first because it's less costly, quicker, and requires fewer formalities.
- The **provisional application** provides flexibility, as any improvements to the invention after filing can be added in the final application. This means the inventor doesn't need to have all the invention details finalized or all experimental data completed. Filing a provisional application also secures an **early priority date**, which is important in establishing the patent rights.

PATENT APPLICATION FORMS

- According to the provisions outlined in the **Patent Act of 1970 (Section 39)** and **the Patents Rules of 2003 (Rule 7, 54, 135, and sub-rule (1) of rule 20)**, the patent application submission involves **utilizing Form-1 and Form-2**.
- **Form-1 solicits general information** such as the Application Title, Names of Applicant(s) and Inventor(s), and the Type of Application (Ordinary, Convention, PCT-NP (PCT- National Phase), Divisional, Patent of Addition, etc.).
- **Form-2 gathers technical details** and the applicant's choice between filing a provisional or complete application.
- For a '**Provisional Application**,' only the '**Description of the Invention**' and the '**Abstract**' are required.
- In the case of a '**Complete Application**,' additional components include the '**Claims**' and the **manner in which the invention is executed**.

- In a patent, "**claims**" are statements that define **exactly what part of the invention the patent protects**. They set the **boundaries of the invention**—like a fence marking property lines—so others know what they can't use or copy without permission.
- There are two types of claims:
 - a) Independent Claims:** These stand alone and describe the main parts of the invention.
 - b) Dependent Claims:** These build on the independent claims and add specific details or variations.
- claims specify the **legal and technical scope** of what the inventor owns and can enforce through the patent.

Figure 6.2: Form-1 (application for the grant of a patent).

"FORM 1" THE PATENTS ACT 1970 (39 of 1970) and THE PATENTS RULES, 2003 APPLICATION FOR GRANT OF PATENT (See section 7, 54 and 135 and sub-rule (1) of rule 20)		(FOR OFFICE USE ONLY)	
		Application No.	
		Filing date:	
		Amount of Fee paid:	
		CBR No:	
		Signature:	
1. APPLICANT'S REFERENCE / IDENTIFICATION NO. (AS ALLOTTED BY OFFICE)			
2. TYPE OF APPLICATION [Please tick () at the appropriate category]			
Ordinary ()		Convention ()	
Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()
PCT-NP ()			
Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()
3A APPLICANT(S)			
Name in Full	Nationality	Country of Residence	Address of the Applicant
		House No.	
		Street	
		City	
		State	
		Country	
		Pin code	
3B CATEGORY OF APPLICANT [Please tick () at the appropriate category]			
Natural Person ()	Other than Natural Person		
	Small Entity ()	Start-up ()	Others ()

4. INVENTOR(S) [Please tick () at the appropriate category]

Are all the inventor(s) same as the applicant(s) named above?	Yes ()	No ()
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If "No", furnish the details of the inventor(s)

Name in Full	Nationality	Country of Residence	Address of the Inventor
			House No.
			Street
			City
			State
			Country
			Pin code

5. TITLE OF THE INVENTION

6. AUTHORISED REGISTERED PATENT AGENT(S)	IN/PA No.	
	Name	
7. ADDRESS FOR SERVICE OF APPLICANT IN - INDIA	Mobile No.	
	Name	
	Postal Address	
	Telephone No.	
	Mobile No.	
	Fax No.	
	E-mail ID	

8. IN CASE OF APPLICATION CLAIMING PRIORITY OF APPLICATION FILED IN CONVENTION COUNTRY, PARTICULARS OF CONVENTION APPLICATION

Country	Application number	Filing date	Name of the applicant	Title of the invention	IPC (as classified in the convention country)

9. IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)

International application number	International filing date

10. IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION

Original (first) application No.	Date of filing of original (first) application

11. IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT

Main application/patent No.	Date of filing of main application
-----------------------------	------------------------------------

12. DECLARATIONS**(i) Declaration by the inventor(s)**

(a) (If the applicant is an assignee: the inventors) can affix their signatures below, or the applicant can choose to upload or enclose the assignment with this patent application, or alternatively, send the assignment via post or electronic transmission, duly authenticated within the specified timeframe). I/We, the inventor(s) named above, confirm that we are the bona fide and original inventor(s) of this Invention, and state that the applicant(s) mentioned here are either our assignee(s) or legal representative(s).

(a) Date

(b) Signature(s)

(c) Name(s)

(ii) Declaration by the applicant(s) in the convention country

((If the applicant in India differs from the applicant in the convention country: the applicant in the convention country can provide their signature below, or the applicant in India can opt to upload the assignment from the applicant in the convention country, or include the assignment with this patent application, or send the assignment via post/electronic transmission, duly authenticated within the stipulated timeframe). I/we, the applicant(s) in the convention country, affirm that the applicant(s) mentioned here are either our assignee(s) or legal representative(s).

(a) Date

(b) Signature(s)

(c) Name(s) of the signatory

Source: <http://www.ipindia.nic.in>

Figure 6.3: Form-2 (provisional/complete specifications).

FORM 2 THE PATENT ACT 1970 (39 of 1970) & The Patents Rules, 2003 PROVISIONAL/COMPLETE SPECIFICATION (See section 10 and rule 13)	
1. TITLE OF THE INVENTION	
2. APPLICANT(S) (a) NAME: (b) NATIONALITY: (c) ADDRESS:	
3. PREAMBLE TO THE DESCRIPTION	
PROVISIONAL The following specification describes the invention.	COMPLETE The following specification particularly describes the invention and the manner in which it is to be performed.
4. DESCRIPTION (Description shall start from next page)	
5. CLAIMS (not applicable for provisional specification. Claims should start with the preamble — 'I/we claim' on separate page)	
6. DATE AND SIGNATURE (to be given at the end of last page of specification)	
7. ABSTRACT OF THE INVENTION (to be given along with complete specification on separate page)	
Note: - Repeat boxes in case of more than one entry. To be signed by the applicant(s) or by authorized registered patent agent. Name of the applicant should be given in full, family name in the beginning. Complete address of the applicant should be given stating the postal index no. /code, state and country. Strike out the column which is/are not applicable	

Source: <http://www.ipindia.nic.in>

JURISDICTION OF FILING PATENT APPLICATION

- India maintains **four patent filing offices** where patent applications can be submitted. The choice of filing location is contingent upon the applicant's residence, domicile, place of business, or the origin of the invention. These designated offices are referred to as patent filing jurisdictions.

Table 6.1: Jurisdiction to file a patent in India.

Region	States	Address
Northern	Haryana, Himachal Pradesh, Punjab, Rajasthan, Uttar Pradesh, Uttarakhand, Delhi and the Union Territory of Chandigarh, Jammu and Kashmir and Ladakh.	Intellectual Property Office Building Plot No. 32, Sector 14, Dwarka, New Delhi-110078 Phone: 011-28032491 Fax: 011-28034301 Email: delhi-patent@nic.in
Southern	Andhra Pradesh, Karnataka, Kerala, Tamil Nadu, Telangana and the Union Territories of Pondicherry and Lakshadweep	patent Office Intellectual Property Building G.S.T. Road, Guindy, Chennai-600032 Phone: 044-22505242 Fax: 044-22502066 Email: chennaipatent@nic.in

Western	Maharashtra, Gujarat, Madhya Pradesh, Goa and Chhattisgarh and the Union Territories of Daman and Diu & Dadra and Nagar Haveli	BoudhikSampada Bhawan, Antop Hill, S. M. Road, Mumbai - 400 037. Phone: 022- 24153651, 24148165 Fax: 022-24130387 Email: mumbaipatent@nic.in
Rest of India	Remaining States	Intellectual Property Office Building, CP-2 Sector V, Salt Lake City Kolkata-700091 Phone: 033-23679101, 033-23671987 Fax: 033-23671988 Email: kolkatapatient@nic.in

Source: <http://www.ipindia.nic.in/jurisdiction-of-patent-offices.htm>

Publication

- Once the patent application has been filed at the Regional Patent Office, the **patent application is kept secret for 18 months in the Patent Office.**
- **After the expiry of 18 months** (from the date of filing of the application or the priority claimed date, whichever is earlier), the **application is published in the Official Journal of Patent Office.**
(<https://ipindia.gov.in/>)
(<https://search.ipindia.gov.in/IPOJournal/Journal/Patent>)
- The purpose of publishing the application is to **inform the public about the invention.** The **publication of an application is a mandatory step.**

PRE-GRANT OPPOSITION

- **Pre-Grant Opposition** is a process where someone can object to a patent application **before** it is officially granted.
- **Objection Period:** If anyone believes that the invention described in the patent application doesn't deserve a patent, they can **challenge it within 6 months** after the application is published.
- **Outcome of Objection:** If there's an objection, the patent office will review it. The application could be **rejected** if the objection is valid, or **approved for further examination** if the objection is dismissed.
- **Publication Timeline:** Normally, patent applications are **kept confidential for 18 months**. However, if the applicant needs to sell, license, or find investors sooner, they can ask to make it public earlier.
- **Form-9:** To make the application public before 18 months, the applicant fills out **Form-9** and submits it to the patent office.

- Competitors often monitor **patent databases** (like the USPTO, WIPO, or their national patent office databases) to track new patents in their industry. This allows them to see newly published applications that might impact their business.

EXAMINATION

- **Patent Examination** is a crucial step to ensure that an invention qualifies for a patent. Here's how it works, simplified:

1) Examination Process: After an inventor submits a patent application, the invention goes through a thorough check by a patent examiner. The examiner reviews whether it meets key requirements, like being **new** (not already known) and **inventive** (not an obvious improvement).

2) Examiner's Queries: Often, the examiner has questions or needs clarification about parts of the invention. They send these queries to the inventor, who must respond with explanations or adjustments. If the examiner is satisfied with the responses, they approve the application to move forward for a patent grant.

3) Request for Examination: An important part of the process is that **patent applications don't get examined automatically**. After the application is published, the inventor must **request an examination** by **filling out a form (Form-18A) within 48 months (4 years) of filing the patent application**. This starts the official examination process.

Example: Imagine an inventor applies for a patent on a **self-cooling water bottle**. Once they file the patent application, they need to wait for it to be published. After publication, they submit a request for examination.

The examiner then checks if the bottle is truly unique (not already made by others) and inventive (not an obvious improvement on existing bottles). If the examiner has questions, like how the cooling system works, they ask the inventor to clarify. Once the examiner's doubts are cleared, they approve the patent.

GRANT OF PATENT

- **Final Approval:** After the inventor meets all the requirements and answers any questions or objections from the patent examiner and the public, the patent is officially granted.
- **Publication in the Patent Journal:** Once granted, the patent is published in the **Official Journal of the Patent Office**, which is **released every Friday**. This journal shares important patent updates and information with the public.
- **What the Journal Includes:** It lists **new patents granted, updates on pending applications, patents restored after expiration, and notices from the Patent Office**. It also includes **a list of patents that are no longer in use**.

VALIDITY OF PATENT PROTECTION

- **Patent Duration:** A patent gives the inventor exclusive rights over their invention for **20 years** from the date they file the application. This means no one else can use or sell the invention during this time without permission.
- **Renewal Fee:** In India, after a patent is granted, the inventor must **renew it every year** by paying a fee, according to specific rules (**Section 53, Rule 80**). If the fee isn't paid, the patent may be canceled, and the inventor would lose protection.
- **Possible Extensions:** **In some countries, the patent protection period can go beyond 20 years.** This happens because sometimes it takes a long time for the product to get regulatory approval (e.g., from health or safety authorities), which delays the inventor's ability to sell it. An extension helps give the inventor more time to benefit from their invention.

POST GRANT OPPOSITION

- **Post Grant Opposition** is a process where a granted patent can still be challenged. Even after a patent is approved, anyone has one year from the publication date to question or oppose it if they believe it shouldn't have been granted. This challenge can be made through the Patent Office or a court.
- If the patent is challenged successfully, it can be **revoked or canceled**. Here are the main reasons a patent might be opposed:

- 1) **Wrongful Ownership:** If someone believes the applicant didn't actually invent the idea or stole part of it, they can oppose the patent.

Example: If a company employee invents something but the company tries to patent it without crediting them, the employee could challenge the patent.

- 2) **Already Published:** If the invention was already published somewhere before the patent's priority date, the patent can be opposed.

Example: If someone already shared the invention in a research paper before the patent application was filed, anyone can argue it's not new.

- 3) **Public Knowledge or Use:** If the invention was already known or used publicly before the filing date, it's not new and can be opposed.

Example: If an inventor tries to patent a type of tool that's been sold publicly for years, someone could oppose it, claiming it's already widely known.

- 4) Lack of Inventive Step:** If the invention seems obvious and doesn't add anything truly new, it can be opposed for lacking originality.

Example: If an inventor tries to patent a slight variation of an existing invention that doesn't add much value, it can be opposed.

- 5) Non-Patentable Subject:** If the invention doesn't qualify as patentable (according to the Patent Act), it can be challenged.

Example: Certain things, like natural laws or simple ideas, can't be patented. If someone tries to patent a naturally occurring process, it could be opposed.

- 6) Insufficient Details:** If the patent application doesn't clearly describe how the invention works, making it hard to understand or recreate, it can be opposed.

Example: If the description in the patent application is too vague or lacks essential details, anyone can argue it shouldn't have been granted.

COMMERCIALIZATION OF A PATENT

1. **Patent Licensing:** When someone owns a patent, they can allow others to **use, make, or sell** the invention by **granting a license**. This is done through an agreement with terms both parties agree on.
2. **Reasons for Licensing:**
 - The patent owner may already have a main job, like being a professor, and doesn't have time to develop or sell the invention themselves.
 - The owner might not have the **equipment or resources** to manufacture the invention.
 - The owner's facility may not meet **market demand** for the invention.
 - The owner might only want to focus on one location and would prefer someone else to manage the invention in other regions.

3. **Rights of the Patent Owner:** After a patent is granted, the **patent owner (patentee)** has the **exclusive right** to use, license, or sell the invention for up to **20 years**. They must report the patent's use or commercialization each year (this is known as "Working/Licensing of the Patent").
4. **Types of Licensing:**
 - **Exclusive License:** The patent is licensed to only **one person or organization**, who then has sole rights to use the patent for a set time period. No one else, including the owner, can use the invention during this time.
 - **Non-Exclusive License:** The patent can be licensed to **many people or organizations**. The patent owner can grant licenses to multiple parties to make or sell the invention.
5. **Compulsory Licensing:** If the patent owner does not commercialize (or bring to market) the invention within **three years**, anyone can ask the **Controller of Patents** to give them a license to use it. This happens if:
 - The public's need for the invention isn't being met.
 - The invention is **too expensive** for the public.
 - The invention is **not available in India**.

NEED FOR PATENT ATTORNEY / AGENT

1. **Filing Without an Attorney:** Technically, someone can write and file a patent application on their own, without needing a patent attorney or agent.
2. **Why Seek Help from an Attorney:** Even though it's possible to file without help, patent applications are **complex**. It's usually a good idea to work with a **patent attorney** or agent who has the expertise to handle the details and legal language properly.
3. **Legal Requirement for Foreign Applicants:** In some countries, if you're applying for a patent and you don't live or do business in that country, you **must** hire a patent attorney or agent who **lives and works** in that country. This ensures the application follows local rules and is handled correctly.

CAN A WORLDWIDE PATENT BE OBTAINED?

1. **No World Patent:** There is no such thing as a "universal" or "worldwide" patent. **Patents are only valid in the country where they are granted**, meaning you can't get a single patent that protects your invention everywhere in the world.
2. **Filing in One Country:** To protect your invention, you must apply for a patent in the **specific country** where you want protection. If you want patents in many countries, it can be **expensive and time-consuming** because you have to file separate applications in each country.
3. **Regional Patent Offices:** To make things easier, some regions have **regional patent offices** that allow you to apply for protection in multiple countries at once. Examples are:
 - The **European Patent Office (EPO)** covers multiple European countries.
 - The **African Regional Intellectual Property Organization (ARIPO)** covers several African countries.

With these regional offices, you can file **one application** to get protection in all the member countries of that region.

4. **International Patent (PCT):** If you want patent protection in many countries around the world, you can file an application under the **Patent Cooperation Treaty (PCT)**. This allows you to apply for protection in over 190 countries with a single application, saving time and effort.
5. **PCT Membership:** For the PCT to work, your country must be a member. For example, **India** is a member, so Indian inventors can file a single PCT application to seek patent protection in many countries.

DO I NEED FIRST TO FILE A PATENT IN INDIA

1. Patent Filing in India:

- Generally, Indian residents must first file their patent application in India.
- After that, they can file for patent protection in other countries.

2. Prior Approval Requirement:

- Indian residents usually need permission from the Indian Patent Office before applying for a patent in another country.

However, there are **exceptions** where this approval is not needed:

- **Non-Indian Applicants:** If the applicant is not an Indian resident, they don't need prior approval from India.
- **6 Weeks Have Passed:** If an Indian resident has filed a patent in India, they can apply in other countries after 6 weeks without needing approval.

- **Foreign Collaborations:** If the invention involves two or more inventors from different countries, and one of the inventors is an Indian resident, but the invention is not for the Indian market, the Indian inventor can file in foreign countries after seeking permission from India (called Foreign Filing Permission, or FFP).
- **International Collaboration:** If part of the invention came from India and the inventor is an Indian resident, they must get approval from the Indian Patent Office before filing abroad.
- **Special Inventions:** For inventions related to **defense, atomic energy**, or certain **utility models**, the inventor must seek permission from the Indian Patent Office before filing in any country.

Example:

- **Example 1 (Non-Indian Applicant):** A company based in the USA invents a new type of phone battery. They don't need permission from India to file for a patent in the USA or other countries, because they are not Indian residents.
- **Example 2 (Foreign Collaboration):** An Indian inventor collaborates with a team in Japan to create a new type of solar panel. Since the invention is not aimed at the Indian market, the Indian inventor must ask the Indian Patent Office for permission before filing the patent abroad.
- **Example 3 (Special Domain):** An inventor in India develops a new technology for **nuclear energy**. Since the invention is related to atomic energy, they must get special permission from the Indian Patent Office to file the patent anywhere.

Patent related forms

- There are over 30 patent-related forms. Some of them are mentioned here.

Table 6.2 : List of important patent application forms.

Form No.	Title of Form
1	Application for a grant of a patent
2	Provisional/Complete specifications
7	Notice of opposition on grant of a patent
7A	For filing a representation opposing grant of a patent
17	Application for compulsory license
18	Request for examination of the application for patent
21	Request for termination of compulsory license
22	Application for registration of patent agent
27	Statement regarding the working of the patented invention on a commercial scale in India
30	Miscellaneous form to be used when no other form is prescribed

Source: http://www.ipindia.nic.in/writereaddata/Portal/IPORule/1_70_1_The-Patents-Rules-2003-Updated-till-23-June-2017.pdf

FEES STRUCTURE

Table 6.3 Fee for obtaining a patent via electronic filing.

Item	Natural person/ startup (₹)	Small entity alone or with a natural person /startup (₹)	Others alone or with natural person/ startup/ small entity (₹)
Provisional/Complete Specifications	1,600	4,000	8,000
Request for Early Publication	2,500	6,250	12,500
Request for Examination	4,000	10,000	20,000
Express Request For Examination	5,600	14,000	28,000
Renewal Fees (Annually)			
3 rd to 6 th Year	800	2,000	4,000
6 th to 10 th Year	2,400	6,000	12,000
11 th to 15 th Year	4,800	12,000	24,000
16 th to 20 th year	8,000	20,000	40,000

Source: http://www.ipindia.nic.in/writereaddata/Portal/IPOFormUpload/1_11_1/Fees.pdf

Types of Patent Applications

- **Provisional Application:** Filed when the invention is not finalized, allowing the inventor to secure a priority date for future claims.
- **Ordinary Application:** Filed with full details and claims, but does not claim a priority date.
- **PCT Application:** An international patent application that allows inventors to seek protection in all countries that are members of the Patent Cooperation Treaty.
- **Divisional Application:** Filed when an original application includes multiple inventions. It splits the application into two or more separate applications.
- **Patent of Addition Application:** Filed for modifications to an invention already patented, without requiring a new renewal fee. It expires with the main patent.
- **Convention Application** - If a patent application has been filed in the Indian Patent Office and the applicant wishes to file the same invention in the one or more Convention countries (e.g. Paris Convention) by claiming the same priority date on which application was filed in India, such an application is known as Convention Application. The applicant has to file Convention Application within 12 months from the date of filing in India to claim the same priority date.

Table 6.18: Commonly used terms in the domain of patenting.

S. No.	Term	Definition
1.	Inventor	Creator of an invention
2.	Applicant	Organization/individual/industry that files a patent application or applies for a patent
3.	Patentee	A person/organization who owns the patent (granted)
4.	Licensee	Organization/individual/industry which obtains a license of the patent from the Patentee for commercialization purpose

5.	Assignee	A person in whose name patent has been assigned legally
6.	In force	The applicant is paying the annuity (renewal fee) for the patent to keep it alive (Active Patent)
7.	Working of a Patent	The selling of a patent to an individual/party for commercial exploitation is called as working of a patent
8.	Patent Specification	Patent specification is a written description of the invention and the way of representation and process of making and using the same
9.	Priority Right	A 'Priority Right' or 'Right of Priority' is a time-limited right, activated by the first filing of an application for a patent
10.	Priority Date	The claimed date on which the first application for the invention is filed

11.	Patent Claims	Claims can be defined as the scope of the protection conferred by a patent, or the protection sought in a patent application. The purpose of the claims is to define which subject matter is protected by the patent
12.	National Phase Application	An application filed to obtain patents in different countries simultaneously based on a single International/PCT application
13.	Patent Revocation	The revocation means cancellation of the patent due to certain reasons, such as lack of patentability or wrongfully obtaining a patent
14.	Restoration of Patent	Once a patent has been ceased (e.g. due to non-payment of the fee) it can be restored within a permitted period by paying the requisite fee

NATIONAL BODIES DEALING WITH PATENT AFFAIRS

Several departments, organizations, and bodies are involved in different aspects of patents, including the Indian Patent Office (IPO), Department for Promotion of Industry and Internal Trade (DPIIT), Technology Information, Forecasting and Assessment Council (TIFAC), and National Research Development Corporation (NRDC).

UTILITY MODELS

- In some cases, a new invention is just a small improvement over existing products. While this improvement may add value, it might not meet the strict requirements for a standard patent, such as Novelty and Non-obviousness. However, some countries offer legal protection for these minor innovations, calling them "Utility Models," "Petty Patents," or "Innovation Patents."
- In Utility Models, the standards for Novelty and Non-obviousness are generally lower or even removed, but the invention must still have practical applications, just like a standard patent. Utility Models are particularly useful for Micro, Small, and Medium Enterprises (MSMEs), as they are quicker and less expensive to obtain compared to traditional patents.
- Since MSMEs often lack the resources for extensive R&D, Utility Models provide an affordable way to protect their inventions that improve existing products or processes and enhance profitability. Unlike patents, Utility Models have a shorter lifespan, typically between 7-15 years, depending on the country.

- Currently, around 80 countries, including France, Germany, Japan, South Korea, China, Finland, Russia, and Spain, recognize Utility Models. However, India does not yet offer this type of IP protection. If India were to adopt Utility Models, it could significantly increase the number of patents filed and granted annually.
- **Example:** Suppose a small company invents a new type of pen that allows smoother writing due to a slight change in the ink flow mechanism. This improvement might not qualify for a full patent due to its limited novelty but could still be eligible for a Utility Model. With this protection, the company could prevent others from copying the improved pen design for a certain number of years without the high cost and long process associated with a standard patent.